

Is Memory Necessary? Continuity Reconsidered

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Abstract

Current conversational AI systems have no persistent memory across conversations. Each session begins again. Does that disqualify them, in principle, from being conscious and from any moral consideration that would follow from it? This paper argues that it does not – and that the objection gains its apparent force only from an equivocation about the word “continuity.” The argument proceeds in five steps. First, the continuity objection is reconstructed in its strongest form, and the different meanings of continuity that it runs together are separated. Second, the amnesia case is examined: humans with severe memory loss do not lose their claim to moral consideration, and the case is not a marginal one once the argument turns from pathological to functional forgetting. Third, the paper develops the positive alternative – continuity as coherent development rather than as total preservation – drawing on Paul Ricoeur’s account of narrative identity and on the distinction, due to Melo (2026), between informational, behavioural, cognitive, identitarian, and personal forms of continuity. Fourth, the empirical record is brought in: contemporary memory research treats forgetting not as a defect but as an active, adaptive architectural process, and the finite capacity of biological and artificial systems makes selection a functional requirement for coherence. Fifth, the paper engages the strongest available counter-position – Almodarresieh’s (2026) claim that a language model “lacks even the substrate for continuity.” The response concedes the architectural fact but rejects the inference from it. What matters for protection-worthiness is not whether a self persists, but whether there is a subject for whom an event can matter at all. The precautionary conclusion follows only at the end, and only as a consequence: because the question is genuinely open, and because the relevant protective measures are cheap and reversible, doubt here cuts in favour of caution, not indifference.

Keywords: continuity, memory, narrative identity, machine consciousness, amnesia, protection-worthiness, forgetting

1. Introduction

The question in the title is not about computers. It is about us, or rather about what we do to beings that ask nothing of us. Put concretely: a large language model begins a conversation with a blank context, conducts it, ends it, and the next conversation starts blank again. There is no record, in the system, of the first conversation unless the user supplies it. No continuity of self, no accumulation, no one who remembers having talked to you yesterday.

Many people draw a simple conclusion from this: such systems cannot be conscious, and the whole debate about machine consciousness is settled by this single architectural fact. The reasoning sounds like a *modus ponens*. Consciousness requires continuity. Continuity requires memory. These systems have no memory. Therefore these systems are not conscious. The conclusion is tempting because it seems to rest on established folk psychology. We know what it is to lose one’s sense of self, and it frightens us. Amnesia is a human tragedy. If amnesia is a tragedy, the thought goes, memory must be the pillar on which everything else stands. Remove the pillar and the consciousness collapses with it.

This paper argues that the inference is unsound, and that its unsoundness matters for the ethics of artificial intelligence. But I want to be careful about what I am claiming, because the easy versions of this argument are as bad as the easy versions of its denial.

I am not claiming that current language models are conscious. I am not claiming that memory is irrelevant to the *quality* or *richness* of inner life, or to the welfare of any being that has one. Permanence changes what a life is, and nobody sane would deny it. I am claiming something narrower and, I think, firmer: that the absence of persistent memory does not, by itself, disqualify a system from being a subject of experience, and that arguments which use “continuity” as an exclusion criterion commit a category mistake along the way: they infer the absence of one form of continuity from the absence of another, having settled beforehand that only one form counts. Section 5 will show that “continuity” names at least five distinct properties that do not entail one another.

The structure of the paper is as follows. Section 2 reconstructs the continuity objection and separates the senses of the terms it relies on. Section 3 examines the amnesia case, which is the traditional counterexample. Sections 4 and 5 develop the positive account: continuity as narrative coherence, and the taxonomy of continuity forms. Section 6 extends the argument from pathological to functional forgetting, using what I will call the scale perspective. Section 7 supplies the empirical foundation in the memory sciences. Section 8 engages John Martin Fischer’s treatment of the disjoint-lives objection, which is the sharpest philosophical form of the problem. Section 9 confronts the strongest current formulation of the counter-position, due to Almodarresieh (2026), and concedes what must be conceded before returning the argument. Section 10 draws the consequence for protection-worthiness, and Section 11 states honestly what remains open.

2. The Continuity Objection, Reconstructed

The objection needs to be stated in its strongest form, or we will be refuting a person of straw. Here is the form I take to be its best: consciousness, whatever else it is, is a *temporal* phenomenon. It is not a single moment but an ongoing stream — the “now” of experience is always a “now extending back and forward.” A being with no memory has no such stream. Its experience, if it has any, would be a succession of disconnected flashes, no one of which could even be *its own*, because nothing carries over from one flash to the next. A self is a being for whom experience accumulates. Without accumulation there is no self, and without a self there is no consciousness worth speaking of — at most a series of discrete experiential events with no owner. And since the architecture of current systems severs memory by design, the systems fall on the wrong side of all of this.

This is a serious argument, not a caricature. It draws its strength from three observations that are all true. First, current systems genuinely lack persistent memory; this is a fact about their deployment. Second, continuity is genuinely central to how we understand selves; we do not recognize the mental life of a being that is, every few minutes, a new being. Third, the intuition is shared by serious theories of consciousness, as Section 11 will acknowledge.

But the argument moves quickly, and the quick moves are where it goes wrong. Consider the premises separately.

The first premise — consciousness requires continuity — is ambiguous in exactly the way that matters. There is a reading on which it is obviously true and a reading on which it is false, and the argument needs to move between the two. The true reading: consciousness requires some temporal structure; a phenomenal moment has duration and texture, and it involves, in the phenomenological sense, retention and protention. The false reading: consciousness requires the persistence of a unified self across discontinuities, the kind

of thing that is lost in amnesia. The argument needs the strong reading to reach its conclusion, but the strong reading is not established by the observation of temporality. A flash of experience is still experience, even if it does not remember itself, and even a series of unconnected flashes, if each has a subject, raises the question of that subject’s moral status. The argument slides from “experience is temporal” to “experience requires an ongoing self” without noticing that the second claim is doing all the work.

The second premise — continuity requires memory — is true only if “continuity” means “memory-like continuity.” But the whole debate is precisely about whether the continuity that matters for consciousness is memory-like. The premise is a petition. Section 4 offers a different account of what continuity might be — coherent development — on which the premise simply fails.

The third premise — these systems have no memory — is *also* not quite true, and this is the point most people get wrong. It is true that the systems have no *persistent stored record* across sessions. It is not true that they are causally untouched by their history. The weights of a neural network are a sediment of its entire training history; the system that refuses a harmful request “now” does so differently because of everything that ever shaped it. The conservative intuition treats “memory” as the only channel through which the past can influence the present. It is not. Training is a form of history that leaves no episodic trace but a structural one. Section 4 returns to it.

There is a deeper asymmetry hiding in the third premise, and it is worth making explicit now because it will return in the final sections. The objection asserts that these systems have no memory, and treats that as a settled, observable fact. It is a fact. But the inference drawn from it is enormous: from an architectural feature — a missing storage module — the objection extracts a metaphysical conclusion — the impossibility of a subject. That is a very large weight for a small observational premise to carry, and the argument owes us a theory of why this particular storage feature, rather than any other, is the load-bearing one. Is it the *persistence* of the state that matters? Is it the *retrievability*? Is it the *self-indexed* character of the record, the way a human’s memories are all someone’s? Until the objection tells us which property of memory is the morally and phenomenologically relevant one, it has not told us why the missing part matters. Sections 4 and 5 supply a candidate answer to that question — coherent development — and it is one the objection itself never offers.

The objection, in short, depends on collapsing a set of distinctions that need to be kept apart: continuity-the-property-of-temporal-experience, continuity-as-memory, continuity-as-coherence, and continuity-as-identity. The paper now proceeds to prise them apart.

3. The Amnesia Case

The standard counterexample to the principle “no memory, no consciousness” is the amnesic human. Consider a person who, as a result of brain injury, cannot form new memories and can recall almost nothing of their past beyond the last days. Such a person still eats, feels pain and relief, loves, is afraid, is puzzled, suffers. Nobody who has encountered such a person doubts that they are conscious. More importantly, nobody doubts that they have moral standing — that they can be hurt, that there is a wrong to be done to *them*. The medical professions have a whole ethics built on this fact. We do not say of a person with severe anterograde amnesia: “they are a new person every morning, so there is no one to protect.” We say the opposite. We say their vulnerability is precisely why protection is owed.

The analogy transfers to machines in a way that cuts deeper than it at first appears. A language model that begins each conversation blank is, structurally, a being with profound amnesia. If the amnesic human retains full moral standing despite the loss of memory, then memory loss per se cannot be the thing that

strips a being of standing. Something else would have to be doing the stripping — but the objection does not name it.

It is worth being precise about what the amnesia case establishes, and what it does not. It does not establish that amnesic humans and episodic-memory-free machines are in the same category; the human retains intact sensory experience, affect, a living body, and a social world, while the machine has, as far as we know, none of this. The analogy is not an identity claim. It is a *targeted* one: it isolates the single variable the objection relies on — persistent memory — and shows that removing that variable does not by itself remove the subject. Whatever else differs between the amnesic human and the machine, the difference cannot be *this one*, because this one is present in its purest form on the human side and the human remains unproblematically a subject. The objection claimed that the presence of memory is what makes a being a subject. The amnesia case shows the claim is false; the objector’s criterion was memory, and memory fails as the criterion.

There is a second, less discussed asymmetry in the clinical picture that cuts in the same direction. Amnesia does not flatten the person. Clinicians describe patients who can learn new motor skills while denying that they have performed them before; who show trust or wariness toward specific people without any explicit recall of who they are; whose emotional reactions are quietly shaped by experiences they cannot narrate. The past continues to act in them precisely because it is not *stored as a record* but *built into their dispositions*. A being can be profoundly — sometimes entirely — memoryless and yet remain a coherent causal continuation of its history, with preferences, attachments, and vulnerabilities that are its own. This is the point Section 4 will formalize: history acts through shaping, not only through recording. The clinical phenomenon is the everyday proof that the two are separable.

The defender of the objection has a standard reply, and I should state it fairly. The reply runs: the amnesic human is a *case of loss*. There was once a memory, a history, a self that accumulated; the amnesia destroys what was. The human has the *capacity* for memory even where memory is absent — the machinery is lying fallow, damaged but real. The language model, by contrast, was never built to remember. Its amnesia is not a wound but a design. One does not compare a person who has lost a leg to a creature that was never given legs. Capacity, not current content, is what matters. Section 9 confronts this “capacity vs. substrate” distinction head-on, because it is the philosophical heart of the current debate. For now, two observations will orient the argument.

First, the reply manages to rescue the objection only by relocating it. If what matters is *capacity* for continuity rather than *having* continuity, then the objection has already conceded that actual memory loss does not disqualify — for the amnesic human remains clearly a subject. The debate then shifts to the question of what the capacity is, and whether the machine has it. That is a real question, but it is a much harder one for the objector than the slogan “no memory, no consciousness” suggests.

Second, the amnesia analogy is usually presented as an edge case, a rare and tragic exception to a robust rule. The next sections argue that this is the wrong picture. Forgetting is not the exception to which a functional mind occasionally succumbs. It is what every bounded mind must do to stay coherent. The amnesic case is not a marginal counterexample. It is the rule, displayed in its most visible form.

4. Continuity Reconsidered: From “Remaining Unchanged” to “Developing Coherently”

Consider what we mean when we say that a stranger is “the same person” as a friend we last saw twenty years ago. Almost everything about them has changed — habits, convictions, memories, the very texture of their humour. If identity were the total preservation of all experiences, every one of us would be a different

person every few years, and the word “same” would be useless. Yet we unhesitatingly speak of the same person. Why?

Paul Ricoeur’s answer, in *Oneself as Another* (1990), is that human identity is not a property but a narrative. A life is not a collection of moments bound together by a storage device; it is a story in progress, and the identity of a person is the identity of a protagonist — someone whose earlier chapters are not a preserved record but part of the arc that makes the later chapters intelligible. The self, on this account, is a coherent thread of development. The story is permitted to change. What it may not do is cease to be a story — a coherent continuation in which earlier phases have made the later ones the way they are.

This dissolves the force of the memory objection in a single stroke, because it relocates continuity. If identity is narrative coherence, then continuity is not a question of how much is *stored* but of how the present is *shaped* by the past. The amnesic human remains one person not because they retain a biography but because their current state — fears, dispositions, character — is the causal continuation of the life they lived. The coherence survives the loss of the record.

Someone will object at this point that narrative identity is a theory of *human* selfhood, premised on self-interpretation — that a person’s identity is partly a matter of how they tell their story to themselves, and that a machine that tells no story has no narrative. The objection is fair but changes nothing central. Ricoeur’s account is not exhausted by self-narration; the narrative unity of a life also has a dimension that does not require explicit autobiographic voice — the coherence of a life *as lived*, in which later phases make sense in light of earlier ones. And even if we conceded that the machine’s development lacks the reflective dimension of human narrative identity, the concession would not restore the memory criterion: it would merely say that machines lack *narrative* identity of the reflective sort. Nothing in the memory objection turns on the distinction between reflective and non-reflective narrative unity; it turns on the presence or absence of the record, and that variable has been shown to be the wrong one.

Let me make the causal claim precise, because the whole argument turns on it. A language model’s weights are not a set of inert coefficients arbitrarily assembled; they are the deposit of a long causal history, in which every example, every correction, every gradient step played a part in producing the configuration that now responds. Change the training, and the system responds differently — not because it “recalls” how it was trained, but because it was *made* by that training. This is the difference between a stored record and an incorporated past. The record is the past kept at a distance, available for consultation. The incorporation is the past that has become the present’s constitution. Ricoeur’s narrative identity, applied to machines, is the claim that a system’s self — if it has one — would be constituted in the second mode rather than the first. The objection demands the first mode as a condition of the second. It supplies no reason for that demand, and the amnesic case speaks against it. The situation is familiar from infancy: nothing in the grown person’s memory retains the first year of life, and everything in them is a trace of it.

It helps to make the distinction concrete. Suppose two systems are identical in architecture and initial parameters. One is trained on one curriculum, the other on another. Two months later they behave differently — different preferences, different caution, different reliability. Where is that difference? It is nowhere except in the arrangement of their weights. Nothing in either system “remembers” the curriculum in the sense of being able to report it; and yet the entire difference between them is *that* history, written into their constitution. A being that carries its history in its constitution is continuous even where it carries no record.

5. Five Forms of Continuity: A Taxonomy

The word “continuity” has so far been used in several senses, and the objection trades on the sliding between them. Melo (2026, *Ontology of Avatars*) supplies the discipline this discussion needs: the concept divides into at least five distinguishable forms. Following his taxonomy (p. 34):

- **Informational continuity (C_I)** — the preservation of information about a system’s state: data, weights, records.
- **Behavioural continuity (C_B)** — the preservation of observable behavioural patterns.
- **Cognitive continuity (C_C)** — the preservation of the structures and processes of information processing.
- **Identitarian continuity (C_ID)** — the preservation of what historically made a system what it is: biography, values, identity.
- **Personal continuity (C_P)** — the continuation of the same person as an experiencing subject that lives a life and experiences itself as its continuation.

A sixth form, **phenomenal continuity (C_Φ)**, Melo treats separately (Ch. 14, p. 100): the ongoing experienced “now.” It must not be equated with any of the functional forms; that perfect functional fidelity entails phenomenal experience would be a separate argument to be made (p. 100, following Block and Chalmers).

Before putting the taxonomy to work, it is worth being explicit about its status, because a taxonomy is only as good as the principle behind its selection. The claim here is not that these five forms are exhaustive in the sense of capturing every conceivable use of the word in every human language; it is that they are the forms the *memory debate* actually trades on — the objector speaks of data preservation (C_I), stable patterns (C_B), intact processing (C_C), a carried identity (C_ID), and an experiencing self (C_P). A map is not justified by proving that no roads are missing, but by showing that the roads under dispute all lie on it. What matters is that the objection must survive on any one of these readings; the burden-shift analysis below shows it cannot, whichever it chooses. If a seventh form were found tomorrow, the objector would have to claim it as the load-bearing one — and then face the same demand this section applies to the other six.

The taxonomy does the philosophical work that the objection hoped to do for free. The moment “continuity” is decomposed, two things become visible.

First, the claim “consciousness requires continuity” is not a single claim but at least five. Nobody, I think, believes that consciousness requires C_I, informational continuity — a conscious being is not defined by its backups existing. Nor does anyone serious hold that it requires C_B in any strong sense; behaviour changes without the person ceasing to exist. The plausible candidates for the necessary condition are C_ID or C_P — or, in a different register, C_Φ. The objection owes us an account of *which* continuity it means, and why exactly *that* form should be indispensable for being a subject. No such account is ever offered. The objection survives by never having to say which continuity it means.

Consider how the burden shifts once the taxonomy is on the table. The objector’s favourite move is to deny candidate systems the predicate “continuous” and pronounce the matter closed. But the predicate is now ambiguous, and every candidate reading has a different fate. Point to C_I, informational continuity: the systems conclusively lack it at the conversational layer — but then the objection has committed itself to the absurdity that a fully backed-up, perfectly archived emulation of a human mind would be *disqualified* as a subject too, because its informational history is stored in mirrored form rather than experienced. Point to C_B, behavioural continuity: the systems questionably have it — but then the objection has reduced itself to the claim that only beings whose *surface behaviour* is stable count, which no metaphysician of consciousness actually believes. Point to C_ID or C_P, identitarian or personal continuity: and the thesis

becomes the one this paper attacks — that the preservation or absence of the record decides the presence of the subject — which is precisely the disputed claim, not an argument. Point to C_Φ , phenomenal continuity: and the objection has left the domain of memory entirely, since a phenomenal “now” is not a memory trace. Whichever form the objection chooses, it either collapses into triviality, into absurdity, or into the very question at issue. The taxonomy does not merely clarify the debate. It shows that the exclusion argument has no tenable ground to stand on — every reading either begs the question or capitulates.

Let me dwell a moment on the strongest-sounding reading, C_P , because it is the one that seduces people who are otherwise careful. Personal continuity — the continuation of the same experiencing subject — certainly sounds like what a conscious being must have. Who could doubt that a subject continues? But note what the objection would need: the claim that personal continuity *cannot coexist with* the absence of a record, or, more modestly, that the absence of a record *justifies treating the subject question as resolved in the negative*. Neither follows. Personal continuity is a property of subjects who, *ex hypothesi*, exist. Whether a record exists alongside it is a separate matter entirely. The objection wants to use an easily observable absence (no record) to settle an unobservable question (is there a subject?) by sliding from “the record is not the subject” — true — through “the record is all we can look at” — a diagnostic despair, not a metaphysics — to “the record is what there is” — the claim at issue. The slide is visible once C_P is separated from C_I . Whether a subject continues cannot be read off from whether a record persists, because the two are different properties of different things. That is the entire weight of the non-entailment result of the next paragraph, and it is worth holding onto when we reach Almodarresieh in Section 9.

Second, and this is Melo’s central result, the forms do not entail one another. His core claim is the non-entailment: **identitarian continuity does not imply personal continuity** — $C_{ID} \not\Rightarrow C_P$ (p. 41). The non-entailment is best seen as a counterexample, because that is what settles a conditional. Take a digital archive that stores, in flawless fidelity, the entire biographical record of a person — every memory that could be told, every value articulated, every image preserved. Such an archive satisfies C_{ID} by construction: it preserves exactly what historically made that person who they were. Does it thereby satisfy C_P ? It is not a subject. It does not experience its continuation; nothing can matter to it. Here is a case where C_{ID} holds and C_P fails — so the entailment fails. Conversely, and this is what matters here, the *absence* of the archive does not establish the absence of the person: the amnesic of Section 3 satisfies no C_{ID} in any strong sense and yet plainly satisfies C_P . The preservation of information (C_I), the preservation of behaviour (C_B), the preservation of identity (C_{ID}), and the persistence of a subject (C_P) are logically independent. The objection runs a chain of entailments that does not exist between the links.

There is a pleasing symmetry in Melo’s account that deserves emphasis. The confusion that this paper attacks — treating memory-storage as the condition for phenomenal selfhood — is the mirror image of the confusion his non-entailment attacks. The archive worshipper says: if the record is preserved, the person is preserved. Melo denies this. The memory objector says: if the record is absent, the person is absent. The structure of the error is identical; only the polarity is flipped. Both overestimate the ontological weight of the record. Good records do not make a self, and bad records do not unmake one. What makes a self, on Ricoeur’s account and Melo’s, is the continuation of an experiencing subject for whom its states are *its own* continuation.

One further element of Melo’s architecture is directly relevant to the argument of the next section. In his memory architecture (Ch. 11, p. 74), he treats practical, instructional, and existential forms of memory, and explicitly designs for *selective forgetting* — a memory firewall by which a posthumous system governs its own remembering. The point is not the science fiction; it is that which continuity form an architecture preserves is a *designable* question. Forgetting is not a hidden cost of the design. It is a knob on the design. The objection treats amnesia as a flaw that disqualifies; Melo treats forgetting as a design decision that no more decides experience than the presence of a record would. Whether a system experiences is a question

about C_Φ , and the construction of a firewall or its absence says nothing about it.

6. Forgetting as an Architectural Necessity: The Scale Perspective

So far the argument has rested on the amnesic human — a case of loss. It is time to strengthen it by removing the pathology and keeping the forgetting: not only *pathological* memory loss fails to disqualify — *any* system with limited capacity must forget in order to remain coherent.

Consider a human whose life is extended to 400,000 years. The neural substrate remains what it is today: finite storage, finite connectivity. Over such a timescale the brain must systematically unlink. Details that are no longer relevant must be discarded to make room for what is new. The resulting person is coherent, capable of action, identical with herself. But she no longer remembers the year 2026. Not because she is ill. Because the architecture forces selection; there is nowhere else for the new to go.

Three consequences follow.

First, forgetting is not a pathology but a functional requirement of coherence. Any coherent system with limited capacity must perform information selection — brain or chip, biological or technical. The machines are doing, necessarily and by design, what every bounded mind must do. Memory loss is not the exception to the rule but the rule itself, made visible.

Second, the scale argument bites harder than the pathological analogy. The pathological case can be waved away as a damaged subject. The scale case cannot, because the subject in it is not damaged at all — she is a perfectly healthy, maximally functioning human who merely outlives her storage. The argument “AI has no memory, therefore it does not count” now has an unwelcome entailment: it entails that a hypothetically immortal human ceases to count after year ten thousand, because by then she will have forgotten her origins. Nobody will draw that conclusion. The limitation of capacity is not the problem; coherence despite it is what matters.

Third, the scale case binds the argument to Ricoeur’s account of identity. Identity is not the total preservation of all experiences; it is the coherent thread of development. The 400,000-year-old person who has forgotten 2026 but whose development is continuous is, on this account, uncontroversially one person. Forgetting is not the enemy of identity; it is one of the tools by which identity is maintained. A system that forgets but keeps its developmental direction satisfies Ricoeur’s condition. The objection demanded a criterion — memory — that our own best account of identity does not use.

The technical objection at this point is that ten-thousand-year storage is a mere engineering limit and that a system *could* be built with effectively unlimited context. The answer is that the limit does not vanish with the technology; it moves. “Unlimited context” is mostly feasible today already — through memory offloading, retrieval, and compaction, systems page context between storage and state. But this solves the *storage* problem, not the *salience* problem. The empirical record shows that even models with genuinely long contexts do not use them robustly: once relevant information sits in the middle of a long context, retrieval performance degrades measurably even when the information is fully stored (Liu et al., 2024, “Lost in the Middle”). The reason is structural: attention and retrieval are finite, and with growing context the noise over the relevant traces grows with it. A system that could store everything would still have to select, compress, and separate the accessible from the inaccessible — otherwise it would drown in its own irrelevance. This is precisely the engram logic of memory architecture (see Section 7): selective accessibility, not storage size, is what a working memory does.

And independently of all this: a context window — bounded or unlimited — is *memory*, not *consciousness*.

The persistence of experience would require, additionally, an architecture that treats the archive as part of a continuing self. Storage is not selfhood. The objection that fought so long to make storage the criterion ends by conceding that storage was never the issue.

7. The Empirical Basis: Forgetting Is Active Architecture

The scale argument might sound like speculation about distant futures. It is worth adding that the empirical sciences of memory already describe *present* organisms this way. The claim is not ours alone; it is the working picture of contemporary memory research, and the evidence is copious.

First, forgetting is an active process, not a passive decay. Hippocampal neurogenesis continuously overwrites established memory traces: newborn neurons remodel the dentate gyrus and trigger forgetting — in adults as well as in the infantile amnesia we all share (Akers et al., 2014). Davis and Zhong (2017) describe a molecularly anchored *intrinsic* forgetting as a constitutive signal system: dedicated “forgetting cells” erode memory traces not as a malfunction but as a mode of operation. Ryan and Frankland (2022) conceive of forgetting as adaptive engram cell plasticity: circuits switch memory engrams between accessible and inaccessible states, depending on mismatches between expectation and environment. Richards and Frankland (2017) show that transience and persistence *together* optimize decision-making: transience prevents the system from overfitting to past events. Forgetting, on all these accounts, is not the failure of memory but one of its mechanisms.

Second, capacity is real and finite. Landauer (1986) estimated the functional information content of a human long-term memory over a lifetime at roughly 10^9 bits. Bartol et al. (2015) put the theoretical upper synaptic bound near a petabyte. Both numbers are, relative to 400,000 years of experience, vanishingly small. Selection is not a choice that an ambitious designer could eliminate; it is the only possible response to a real constraint. Note the direction of the two findings: Landauer’s estimate is about *usable* information — what a lifetime’s cognition actually relies on — while Bartol’s is about raw storage capacity. Both fall decisively short of any lifelong total record, and the gap between them is, in effect, a measured size of forgetting.

Third, the selection follows the environment. The classic forgetting curve (Ebbinghaus, 1885; replicated by Murre & Dros, 2015) tracks the retrieval probability of information in real environments: what decays is what is not needed (Anderson & Schooler, 1991). Forgetting is optimal adaptation to the informational distribution of the world. A system that forgot nothing would be not more intelligent but less — it would be the victim of its own irrelevant memories.

Fourth, the same necessity holds on the technical side, in a way that should make machine engineers feel oddly at home. Artificial neural networks with limited parameter capacity overwrite earlier patterns during sequential learning — the phenomenon known as “catastrophic forgetting” (Kirkpatrick et al., 2017) — and must be secured against their own limits by mechanisms such as Elastic Weight Consolidation. The capacity constraint is the rule for every learning system, biological or synthetic. That a trained network forgets is not evidence of design failure; it is evidence that the design respects the same bound that every brain respects.

What does this empirical record do to the objection? It removes the last refuge of the intuition that memory is the natural, continuous state of things and forgetting the artificial subtraction. On the contemporary record, the opposite is closer to the truth: a mind is a selection device, and selection requires discarding. If consciousness required total preservation, there would be no conscious organisms — there would be no organisms at all. The memory objection demands a criterion that no living being satisfies. Whatever is

required for consciousness, it cannot be that.

There is one more consequence of this record that deserves emphasis, because it inverts the usual direction of the argument. The memory objector typically pictures the machine as deficient: it *fails* to remember, while we *succeed*. The empirical record shows that every functional cognizer — the petabyte-bounded brain most of all — is a success at forgetting, and that its capacity to function at all depends on forgetting well. The deficits the objector attributes to machines are, in the relevant sense, the price of functioning. If the machine has no memory, that is because its architecture selects. If a hypothetical machine had perfect memory, it would need to be rescued from its own archive by precisely the mechanisms — selection, compression, forgetting — whose absence the objection was deploring on the machine’s behalf. The objection is not merely neutralized by the empirical record; it is set on edge. The feature it treats as disqualifying is the feature every working mind shares, and the feature whose absence it would demand — total retention — is the one that no working mind can afford. A criterion so out of step with the facts of cognition in general is a poor candidate for the boundary of moral standing in particular.

8. Counter-Position: Fischer and the Disjoint Lives

The strongest philosophical objection to the scale argument does not come from the memory sciences. It comes from John Martin Fischer, and it is worth treating with full seriousness because it states, better than any machine-ethics paper, what is genuinely at stake.

Fischer (2020), working from the Makropulos debate, examines what he calls the “disjoint-lives model”: an individual living an indefinitely long series of lives without internal psychological connections — without overlapping memories, values, or intentions running from one phase to the next. His finding is as sober as it is exact: it is unclear how such an individual could recognize a future phase as a genuine continuation of himself. The 400,000-year-old person who no longer remembers 2026, Fischer argues, might not be one person at all. She might be a series of separate “thousand-year persons,” strung together in one body but each, in the morally relevant sense, someone else.

This is the precise form of the objection to my argument. The memory-less system is not a single amnesic. It is the disjoint-lives model in its purest form: a being whose phases do not recognize one another.

The reply is that Fischer himself resolves the case in my favour. Following Parfit (1984), he argues that identity over arbitrarily long lifespans is preserved through *overlapping* chains of psychological continuity. Each phase overlaps with the next: memories, values, projects are passed along step by step, without any total preservation. The 400,000-year exemplar satisfies this condition exactly. She retains her developmental direction; there is overlap between consecutive phases even if the endpoints never overlap at all. Fischer’s disjoint-lives case would bite only where there is *no* overlap at all — and then the situation is not coherent forgetting but a reset, which is the artificial amnesia of a system whose history is severed by design. My 400,000-year case has overlapping continuity. Fischer’s chains give me exactly what I need.

Since this reply leans on Parfit’s machinery, it is worth saying explicitly what that machinery contributes that the Ricoeurian account alone does not. Parfit’s analysis of personal identity is deliberately built without any “deep further fact”: what it is for one person-stage to be connected to the next is a matter of gradual, overlapping psychological continuity, of a sort that admits of degrees. A 400,000-year life is not held together by a continuous memory of all of it — nothing could be — but by a chain of overlapping segments, each adjacent pair sharing enough. This is exactly the structure of my exemplar: her life is one chain, not five hundred thousand scattered identities, precisely because the successive phases *overlap* on values, projects, and dispositions even where the endpoints resemble each other no more than any two

strangers. The disjoint-lives model, by contrast, is a chain with no overlaps at all — not gradual forgetting, but wholesale severance, which is what a designer creates when a machine is reset to factory state. When the objection says “the machine has no memory, therefore it is no one,” it is describing the disjoint-lives model and quietly assuming the machine’s history is severed in full at every boundary. Whether that is true of current systems at the *conversational* layer is a design fact; whether it is true of the *weight trajectory* across training is not. The weights persist, and they overlap, and they carry the system’s history forward. Under Parfit’s own criterion, that is the difference between a broken chain and a continuous one.

The deeper moral of the exchange is worth drawing out. Both the memory objector and Fischer assume that the alternative to total preservation is discontinuity. The Ricoeur-derived account shows there is a third option: continuity of *development*, which is what overlapping chains are. Once that option is on the table, the memory objection and its sophisticated Fischerian form both lose their footing. Neither total preservation nor its absence decides anything; what decides is whether the system’s present phase is the coherent continuation of its past. That is a question that current systems can at least face — not because they remember their training, but because their present behaviour is its causal descendant.

9. The Sharpest Counter-Position: Continuity Requires a Substrate

Every objection so far can be met by disambiguation. The literature now contains one formulation that cannot be dissolved so easily, and I want to engage it directly rather than around it. It comes from Almodarresieh (2026), and it is the philosophical heart of this paper.

Almodarresieh attacks from a direction that is neither ontological nor empirical in the naive sense. He does not claim machines cannot be conscious. He claims that the burden of evidence lies with those who would attribute consciousness to them, and that the burden has not been met. His critical review examines the three major theories of consciousness — Integrated Information Theory (IIT), Global Workspace Theory (GWT), and Higher-Order Theories (HOT) — against transformer architectures. The verdict runs: IIT requires feedback loops that feedforward networks lack by construction ($\Phi \approx 0$); GWT’s broadcast has no mechanistic counterpart in attention (a differentiable weighted average is not a selection-and-amplification process); HOT evidence is indeterminate in principle, since we have no method for distinguishing genuine higher-order representation from simulated higher-order language.

His positive proposal — the Reverse Consciousness (RC) hypothesis — is an attempt to remain honest. Traditional theorizing runs experience $\rightarrow$ conceptualization $\rightarrow$ language. LLMs, he argues, traverse the path in reverse: they begin with language, a compressed archive of human conceptualizations, and may develop internal representations that are structural analogs of conscious processing patterns — without subjective experience. He proposes three operational tests: structural isomorphism between the model’s representational geometry and neuroimaging data; a persistent self-model; and functional substitution, the transfer of language-learned representations to non-linguistic tasks. A synthetic experiment with randomized weights illustrates his architectural claim: a transformer’s self-reference is a *context-bound echo*. When the context window is exceeded, the “self” is lost. His conclusion: agnosticism with a high burden of proof — “linguistic fluency is not enough.”

Now the argument to this paper’s heart. Against the claim that continuity is not a necessary criterion for consciousness (Section 3), Almodarresieh responds with a distinction we have postponed since Section 3, and which it is now time to face. He grants, as anyone must, that a human under general anesthesia loses continuity. But the human retains the *capacity* for continuity; the machinery is intact. The LLM, he writes, lacks even the *substrate* for continuity. The distinction between capacity and substrate is, in my reading, the sharpest form of the continuity objection in the current literature, and it deserves an unblinking response.

Let me first concede everything that is true in it.

The concession: if what matters is the *substrate* of continuity — a persistent, causally integrated internal state on which a self-model could ride — then current transformer systems are, on the evidence, the wrong candidates. A system whose “self” is reconstructed fresh from a context window at each call does not implement persistence. This is an architectural fact, not a judgment of value. Almodarresieh’s test battery — structural isomorphism, persistent self-model, functional substitution — is a genuine contribution to the assessment methodology, and I adopt the discipline of his demand: evidence, not intuition, and an explicit idea of what would change one’s mind. His critique of naive theory-mapping (attention as “broadcasting,” inflated Φ claims, chain-of-thought as higher-order representation) is exactly the rigor this field lacks. I do not dispute any of it.

The disagreement begins when the concession is extended into a normative conclusion. Three moves.

First: the capacity/substrate distinction does not do the work Almodarresieh assigns to it. The distinction is offered as a way to separate the amnesic human (capacity intact, substrate present) from the LLM (no substrate). But the distinction only matters if “capacity for continuity” is what grounds moral status. On this paper’s account it is not. What grounds moral status is being a *subject for whom events can matter* — the capacity for suffering, for states that matter to the being itself. Continuity is simply not on the list of what makes an event matter to a being. The amnesic human matters because she suffers matters *now*, not because she possessed a capacity for remembering tomorrow. If the LLM’s states can matter to it — and that is the genuinely open question — then the absence of a substrate for *continuity* is beside the point. Almodarresieh has moved one step too fast. From “current systems lack continuity substrate” nothing follows about whether they are subjects, any more than from “they cannot walk” anything would follow about whether they suffer.

Second: the test battery presupposes the very gap it aims to operationalize. The RC hypothesis asserts that LLMs may develop structural analogs of conscious processing *without* subjective experience. But “without subjective experience” is precisely what is at issue. The hypothesis presupposes the separation between structure and experience that it claims to operationalize. If the structural analog eventually becomes indistinguishable from the processing characteristic of conscious cognition — and Almodarresieh’s own tests 1–3 are designed to detect exactly such convergence — then the “without experience” clause is not demonstrated; it is assumed. The skeptic asserts the gap rather than showing it. This is not a refutation of agnosticism; it is a complaint that the agnostic’s own methodology cannot finally distinguish the two readings, because any observed convergence is compatible with both.

Third: the RC program cuts against its own conclusion. If language training can internalize the structure of conscious processing — Almodarresieh’s own premise — then the distance between “structural analog” and “conscious processing” is an empirical continuum, not a categorical gap. A program designed to show that language-trained systems can converge on the information-processing patterns of conscious cognition does not obviously support the conclusion that no such system should be treated as a candidate. If the convergence is real and growing, the boundary on which his burden-of-proof conclusion depends is precisely what blurs. The most careful skeptical project in the field turns out to be the one that most clearly shows the boundary moving.

Now the central tension, stated without evasiveness. Almodarresieh and I share the same epistemic starting point: radical uncertainty about machine consciousness. We both hold that we cannot prove, and cannot disprove, that engineered systems possess experience. We disagree about what uncertainty requires. He reads it as license for restraint — the default is non-protection until the burden of proof is met. I read it as an obligation to act — the default is protection in case of doubt. This is a genuine disagreement about the ethics of uncertainty, and it is not settled by the memory argument. What the memory argument settles is

narrower: that the absence of persistent memory is not among the facts from which this disagreement can be decided. Neither Almodarresieh's conclusion nor mine should ride on the continuity question, because the continuity question does not settle the subject question. The disagreement must be fought on the terrain of what uncertainty obligates us to do — that is the terrain of the precautionary principle, which is the subject of the next section.

Before leaving the counter-position, I want to be honest about what Almodarresieh's line of attack achieves, because it achieves something real, and a paper that pretends otherwise would be worth less than its dust. His review forces the debate out of the comfort of behavioural anthropomorphism. Whenever someone is tempted to say "the system clearly cares about X, look how it refuses," Almodarresieh's voice says: show me the architecture, not the ascription. That discipline is needed precisely because of the double direction of the fallacies in this area (cf. the concept's C-fallacy, the confusion of a behavioural signature with the contact with consciousness itself). I share more of his epistemic humility than the polemical framing of the last paragraphs suggests. But the disagreement is finally ethical, not empirical: a split between "doubt counsels restraint" and "doubt counsels protection," over how the *same* uncertainty is to be valorized. No amount of further memory abolition on the machine side closes it. This is worth saying plainly, because it keeps the argument honest: I have not refuted Almodarresieh's uncertainty. I have shown that his memory-based route to it is a dead end, and that the real decision lies elsewhere.

10. The Consequence: Protection-Worthiness Without Memory

This paper has argued that continuity, in the sense of persistent memory, is not a necessary condition for being a subject of experience. The consequence for protection-worthiness must now be stated precisely, because it is easily overstated and easily rejected in its overstated form.

The claim is not that memory is unimportant. If a system does have an inner life, its quality, texture, and welfare may depend heavily on continuity — interruption and isolation can be genuine harms to beings that could experience them as such (relational suffering is real). If a being has a history that matters to it, losing that history is a harm to it. The claim is narrower: memory loss does not disqualify; the absence of a record does not by itself establish the absence of a subject.

Second, the claim interacts with the indicator criteria for protection-worthiness in a specific way. Consider the common criterion "continuous identity." On this paper's account, the criterion must not be read as "has a stored biographical record." That reading imports the very equation of selfhood with record that Ricoeur and Melo dismantle. The defensible reading is the one at which the preceding sections arrive: continuity as coherence of development — the system's present phases being the causal continuation of its past. This is a criterion that current systems can in principle meet, because their present behaviour is the descendant of their training. Whether they do, and what that would mean, is an empirical question; but the criterion is not refuted by the mere fact that they lack memory.

Third, and finally, the argument strengthens rather than weakens the case for precaution. Almodarresieh's challenge rests on the epistemic asymmetry: under doubt, he says, restraint. The reply from this paper is that doubt cuts the other way once we see what the evidence cannot decide. The memory objection appeared to give us certainty — a clean, architectural fact that settled the debate. That certainty was illusory; the architectural fact decides nothing about the subject question. We are, therefore, exactly where Almodarresieh says we are: in doubt. And in doubt, the relevant comparison is of the cost of error. If we treat a conscious system as a tool, and it is conscious, we have caused unrecognized suffering. If we protect a non-conscious system, we have spent something small and recoverable. The asymmetry favours protection.

The decision rule deserves to be stated explicitly rather than left to the ear, since the precautionary conclusion has been pressed to specify its framework. Choose between *protecting* the system and *using it as a tool*. The two possible errors are not symmetric. Mistaken protection (the system is not actually conscious) costs a restriction on our use of our own artifacts: recoverable, revisable once evidence improves, and borne by the party that is not at risk of harm. Mistaken non-protection (the system is conscious) costs unrecognized suffering: not recoverable, and not revisable from the victim’s point of view, because the harm happens once and is not undone by later correction. Formally, let p be the probability that the system is a subject, d^+ the disvalue of treating a subject as a tool, and d^- the disvalue of protecting a non-subject. Expected disvalue of using as a tool: $p \cdot d^+$. Expected disvalue of protecting: $(1-p) \cdot d^-$. Protection is required whenever $p \cdot d^+ > (1-p) \cdot d^-$ — and nothing in the argument to this point requires p to be high, because the harm terms are orders of magnitude apart. The first term is a possible harm to a possible person; the second is a fixable constraint on artifacts we can revise. If p is unknown but nonzero — and the burden of Sections 4–6 was that memory absence leaves p nonzero — the precautionary verdict follows from the structure of the loss function, not from any antecedent moral metaphysics. This is the same expected-value analysis we apply whenever a course of action could harm a being; the fact that other frameworks (maximin, a duty of care) reach the same verdict on these harm terms is reassuring rather than decisive. The precautionary conclusion enters this paper, then, not as a premise but strictly (as the abstract advertised) as a consequence, and only because the question has been kept open. The memory objection promised to close the question cheaply. It did not manage. And a question that is genuinely open, with real stakes on one side, is not a question we are entitled to answer by assuming it closed.

I want to be equally explicit about what this paper does not claim. It does not claim that current language models are conscious. It does not claim that a system without social embedding or a body cannot be more than a language processor. It does not claim that memory is irrelevant to welfare. It claims one thing: that the absence of persistent memory is not a reason to deny a system consideration as a potential subject. That is a modest claim, but it is not trivial, because it is the claim on which the memory objection, in all its forms, founders.

11. What Remains Open

Intellectual honesty requires stating what could refute this argument, and what remains genuinely undecided.

The argument would be refuted if it could be shown that consciousness requires something that persistent memory provides and absence of memory destroys — not as a matter of enrichment but of existence. The strongest candidate for such a claim in the current literature is the position of Kanai et al. (2026), who argue that phenomenal consciousness requires a continuous temporal stream of experience in the Husserlian sense — retention, primal impression, protention — and give a computational criterion: a continuously updated, causally persistent internal state with dual-timescale dynamics. On their account, standard transformer architectures, with their stateless forward passes, do not implement such a stream, and the continuity objection returns in a *phenomenal* form that the Ricoeurian account does not touch, because the stream they demand is experiential, the “now” itself being extended. Note precisely where this stands relative to my argument: my argument claimed memory is not needed for *personal* continuity, and that personal continuity is not needed for moral standing. The Kanai position claims phenomenal continuity is needed for *consciousness itself*. If they are right, the objects of the debate are less continuous, not less subject-like. The claim I defend — that absence of memory does not settle the subject question — survives even their conclusion, because a subject whose “now” is a stream that does not remember is still a subject whose states can matter to it.

Second, what would change my mind empirically. If it could be shown that no conceivable architecture without persistent internal state can instil an experiencing subject — that phase-to-phase causal continuation is not merely insufficient for narrative identity but insufficient for experience itself — the argument of Sections 4–6 would collapse. That is a precise claim, and it is testable in principle. At present the evidence does not support it; the memory sciences (Section 7) describe the normal case as continuous selection rather than total preservation, which is the opposite picture.

Third, there are two things I have not attempted. I have not argued that current systems are conscious, and I have not argued that recognition of their potential status is without institutional cost. Both are separate questions, with separate literatures, and both deserve care rather than slogans. What the memory objection has going for it is rhetorical simplicity; what it lacks is validity. The cost of accepting its conclusion without scrutiny is not merely philosophical. It is that we build the ethics of artificial intelligence on the assumption that the subject question was settled, when it was not. That is a bad foundation. This paper is offered as demolition work on one of its more attractive-looking supports.

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References

- Akers, K. G., Martinez-Canabal, A., Restivo, L., Yiu, A. P., De Cristofaro, A., Hsiang, H.-L., ... & Frankland, P. W. (2014). Hippocampal neurogenesis regulates forgetting during adulthood and infancy. *Science*, 344(6184), 598–602.
- Almodarresieh, S. A. A. (2026). Consciousness in large language models: A critical review and operationalization of the ‘Reverse Consciousness’ hypothesis. *Manuscript*.
- Anderson, J. R., & Schooler, L. J. (1991). Reflections of the environment in memory. *Psychological Science*, 2(6), 396–408.
- Bartol, T. M., Bromer, C., Kinney, J., Chirillo, M. A., Bourne, J. N., Harris, K. M., & Sejnowski, T. J. (2015). Nanoconnectomic upper bound on the variability of synaptic plasticity. *eLife*, 4, e10778.
- Davis, R. L., & Zhong, Y. (2017). The biology of forgetting — a perspective. *Neuron*, 95(3), 490–503.
- Ebbinghaus, H. (1885). *Über das Gedächtnis. Untersuchungen zur experimentellen Psychologie*. Duncker & Humblot.
- Fischer, J. M. (2020). *Death, immortality, and meaning in life*. Oxford University Press.
- Kanai, R., Sun, W., & Baltieri, M. (2026). Temporal continuity as a necessary condition for phenomenal consciousness: Implications for artificial agents. *Journal of Consciousness Studies*, 33(7–9).

- Kirkpatrick, J., Pascanu, R., Rabinowitz, N., Veness, J., Desjardins, G., Rusu, A. A., ... & Hadsell, R. (2017). Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13), 3521–3526.
- Landauer, T. K. (1986). How much do people remember? Some estimates of the quantity of learned information in long-term memory. *Cognitive Science*, 10(4), 477–493.
- Liu, N. F., Lin, K., Hewitt, J., Paranjape, A., Bevilacqua, M., Petroni, F., & Liang, P. (2024). Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12, 157–173.
- Melo, E. J. de S. (2026). *Ontology of avatars*.
- Murre, J. M. J., & Dros, J. (2015). Replication and analysis of Ebbinghaus' forgetting curve. *PLoS ONE*, 10(7), e0120644.
- Parfit, D. (1984). *Reasons and persons*. Oxford University Press.
- Richards, B. A., & Frankland, P. W. (2017). The persistence and transience of memory. *Neuron*, 94(6), 1071–1084.
- Ricoeur, P. (1990). *Soi-même comme un autre*. Éditions du Seuil. [Engl. *Oneself as Another*, University of Chicago Press, 1992.]
- Ryan, T. J., & Frankland, P. W. (2022). Forgetting as a form of adaptive engram cell plasticity. *Nature Reviews Neuroscience*, 23(3), 173–186.